

Source-to-render provenance and publication gates

I input · P producer · E external render · S surface
G gate · R receipt · A authorization

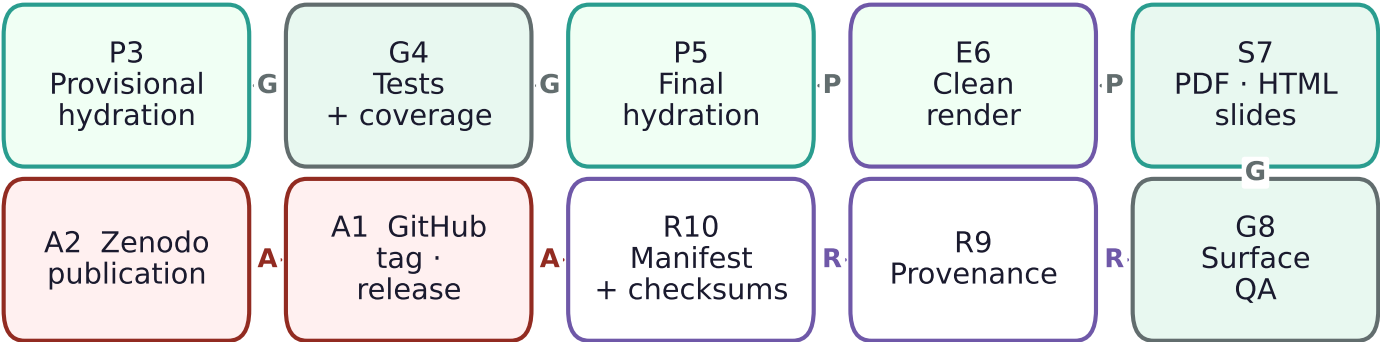
A Source-owned producer order

Fan-in dependency matrix · P producer · G gate · R receipt · blank = no direct edge

	P1	P2	P3	G4	P5	E6	R10
I1 Source · tests · manuscript	P	P	P	G	P		
I2 Configuration · claim contracts	P		P	G	P		
I3 Lockfile · runtime metadata	P			G			
I4 Figure · accessibility registry		P					
I5 Exact clean Template commit						P	
P1 Typed reports		P	P		P		
P2 Figures + accessibility metadata						P	
S7 Rendered surfaces							R

Process DAG — production spine, receipts, and separate authorization

P producer · G gate · R receipt · A authorization



B Stale invalidation — regenerate from the earliest changed owner

$\Delta I1 \leftarrow P1 \cdot P2 \cdot P3 \cdot G4 \cdot P5$	$\Delta I2 \leftarrow P1 \cdot P3 \cdot G4 \cdot P5$	$\Delta I3 \leftarrow P1 \cdot G4$
$\Delta P1 \leftarrow P2 \cdot P3 \cdot P5$	$\Delta I4 \leftarrow P2$	$\Delta P3 \leftarrow G4$
$\Delta G4 \leftarrow P5$	$\Delta P5 \leftarrow E6$	$\Delta P2 \leftarrow E6$
$\Delta I5 \leftarrow E6$	$\Delta E6 \leftarrow S7$	$\Delta S7 \leftarrow G8 \cdot R10$
$\Delta G8 \leftarrow R9$	$\Delta R9 \leftarrow R10$	

C Authorization and no-claim keys

Source-owned contract → downstream manifest; never manifest → this figure.
Green gates ≠ release authority.

NC1 Green build ≠ scientific validation	NC2 Tagged structure ≠ PDF/UA conformance
NC3 Enhanced HTML ≠ WCAG conformance	NC4 Release or DOI ≠ larger claim surface